Project 3

Due: May 1st by the end of the day

Using the code you developed for the laminar boundary layer in project 2, calculate the development of a turbulent boundary layer using the algebraic model described below.

In the inner region, use the Reichardt model. The eddy viscosity is given by

$$\nu_t = \kappa\nu\left[\left(\frac{yu^*}{\nu}\right) - y_a^+ \tanh\left(\frac{yu^*}{\nu y_a^+}\right)\right],$$

Where $y_a^+ = 9.7$.

In the outer region, use the Clauser model that assumes that the eddy viscosity reaches a constant value

$$(\nu_t)_{max} = c_1 U_e \delta(x).$$

$\delta(x)$ is taken to be the location where $u = 0.994U_\infty$. Whenever ν_t calculated from Reichardt's model exceeds $(\nu_t)_{max}$, the Clauser model is used and derivative of ν_t is set to zero. Use κ = 0.41 and c_1 = 0.001. You will need to calculate u^* by taking the numerical derivative of u at the wall.

Make sure you nondimensionalize the expression for ν_t and find the derivatives of these expressions with respect to *y*.

1. Calculate the turbulent boundary layer flow for the same dimensional properties used in the laminar case. However, use a value of $y_{max} = 50\ \delta$ and 250 points in the y direction. Calculate $\delta(x)$ and $c_f(x)$ for each *x* location. Check that the boundary layer grows like
$$\delta(x) = 0.375x(Re_x)^{-1/5}.$$
2. Adjust your results to account for the transition from the assumed initial laminar velocity profile by obtaining a value for x_0, the apparent origin of the turbulent boundary layer.
3. Show that the turbulent boundary layer follows the "law of the wall".
4. Determine the sensitivity of the turbulence model to the choices of κ and c_1.
5. (discussion only) Try increasing the Reynolds number by a factor of 10. What changes are necessary in the grid resolution and stretching factor to obtain a converged solution?
6. (discussion only) Discuss the problems with implementation of this model.

Again, the report should not exceed 5 pages. Appendices should include details of your computer code.

ME 5311: Computer Project #3: Grade Sheet

Item	Score
Description of the problem	/01
Description of the numerical method	/03
Presentation of results	
Velocity profiles (u and v)	/06
Calculation of initial value of *x*	/01
Comparison with boundary layer growth	/01
Comparison with the log of wall	/02
Sensitivity of parameters s, k, c0	/02
Discussion of results	
General description	/01
Accuracy and stability	/01
Appendix – copy of program listing	
Structure	/01
Documentation	/01
Total:	/20